EXPERIMENT 7: DEVELOP A SIMPLE CALCULATOR USING XML-RPC

AIM:

To implement a simple remote calculator using Python XML-RPC client-server communication.

PROCEDURE:

1. Save the server program as xmlrpc_server.py and run it in one terminal.
2. Save the client program as xmlrpc_client.py.
3. Run the client in another terminal on the same computer.
4. Enter two numbers and select an arithmetic operation.
5. The client sends the request to the server and displays the returned result.
6. Make sure that the server is started successfully before launching the client.
7. Test addition, subtraction, multiplication and division with different input values.
8. Test division by zero and invalid operation input to check error handling.
9. Stop the server after completing the client-server tests.

CODE:

```
# xmlrpc_server.py
from xmlrpc.server import SimpleXMLRPCServer

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b
```

```
def divide(a, b):
    if b == 0:
        raise ZeroDivisionError("Cannot divide by zero")
    return a / b

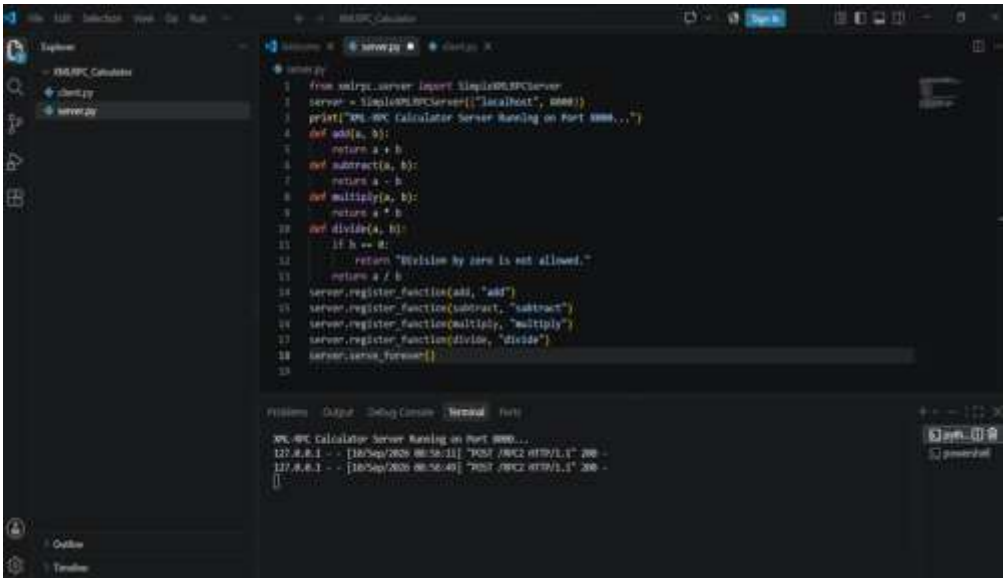
server = SimpleXMLRPCServer(("127.0.0.1", 8000),
allow_none=True)
server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")
print("XML-RPC calculator server running on
127.0.0.1:8000")
server.serve_forever()

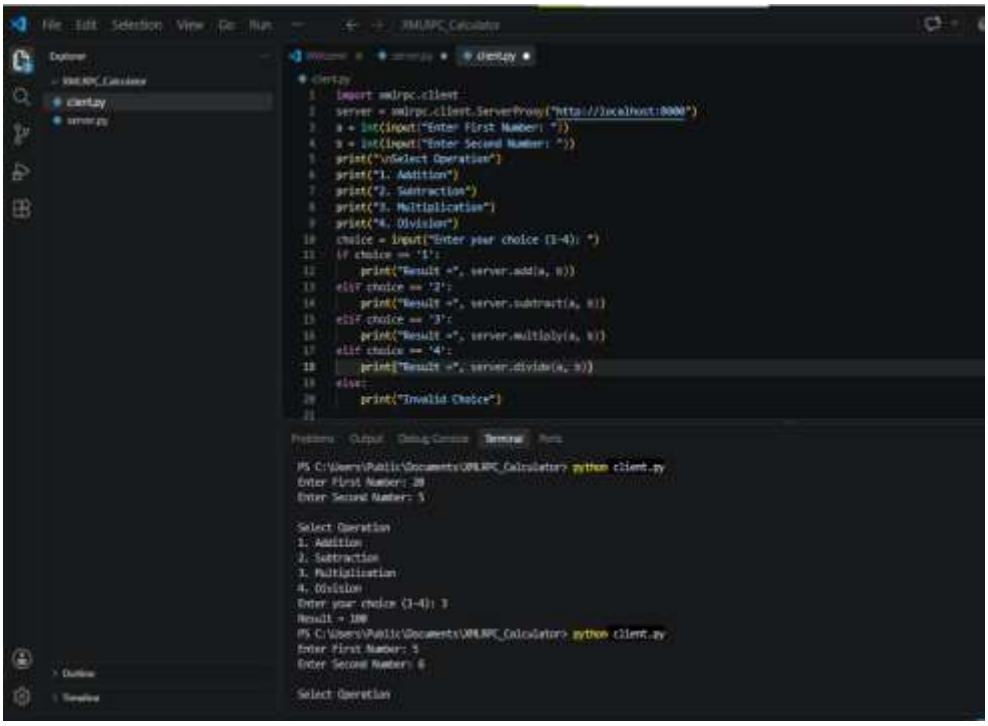
# xmlrpc_client.py (save separately)
import xmlrpc.client

proxy =
xmlrpc.client.ServerProxy("http://127.0.0.1:8000/")
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
op = input("Operation (+, -, *, /): ").strip()

methods = {"+": proxy.add, "-": proxy.subtract,
           "*": proxy.multiply, "/": proxy.divide}
if op not in methods:
    print("Invalid operation")
else:
    try:
        print("Result:", methods[op](a, b))
    except Exception as exc:
        print("Error:", exc)
```

OUTPUT:





RESULT:

Thus, a simple calculator was implemented using XML-RPC client-server communication.